

★ WEEK 4 · PART 1 of 3

ARIMA: Integrating the Trend

Extending ARMA to Non-Stationary Data

Differencing · the order of integration d · ARIMA(p,d,q) · forecasting trends

ARMA needs a stationary series — but most real data trends. ARIMA closes that gap with one idea: difference the series until it is stationary, model the result, then integrate the forecast back.

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PART 1 · ARIMA

From ARMA to ARIMA

The 'I' for integration: how differencing removes a stochastic trend, what the order d means, and how to forecast a trending series honestly.

The Airline That Taught Us to Forecast — Part 1

The dataset that showcased ARIMA

Monthly international airline passengers, 1949–1960, rose relentlessly AND swung in a yearly rhythm that grew with the level — a series no stationary model could touch.

Two problems at once

- A strong upward trend: the mean never sits still.
- A repeating yearly season whose swings grow over time.
- Plain ARMA assumes neither — it needs a stationary series.

Why it became famous

- Box and Jenkins used it as the showcase for their method.
- It motivated both the 'I' (this part) and the seasonal 'S' (Part 2).
- The 'airline model' is still a textbook benchmark today.

The Airline That Taught Us to Forecast — Part 2

The ARIMA idea, and the lesson

The fix was disarmingly simple: difference the series to strip out the trend, model what remains with ARMA, then add the trend back when forecasting.

What ARIMA added

- One integer, d : how many times to difference.
- Everything else is the Week-3 ARMA machinery, reused.
- Trend handled by differencing; season handled next week.

The lesson for us

- Do not fight a trend — difference it away, then model.
- The order d is a modeling choice you must justify.
- This one idea unlocks the vast world of trending data.

The Gap ARMA Leaves

Three reasons ARMA alone is not enough

1

ARMA assumes stationarity

Its whole theory — the ACF, the estimates, the forecasts — rests on a constant mean and variance. Real economic series rarely oblige.

2

Most real series trend

Prices, GDP, sales, and passenger counts drift upward over time. Applying ARMA directly courts the spurious problems of Week 2.

3

Differencing bridges the gap

We already know (Week 2) that differencing removes a stochastic trend. ARIMA simply bolts that step onto ARMA and formalizes it as the order d .

ARIMA in one line: difference to stationarity, model with ARMA, integrate the forecast back.

The 'I' in ARIMA: Integration

Where the third letter comes from

A series is 'integrated' if it must be differenced to become stationary. ARIMA layers exactly that step — the Integration order d — on top of an ordinary ARMA.

The name, decoded

- AR: autoregression on past values (Week 3).
- MA: moving average of past shocks (Week 3).
- I: the number of differences, d , to reach stationarity.

Why 'integrated'

- Differencing is like a derivative; undoing it is integration.
- An $I(1)$ series is the running sum of a stationary one.
- So we difference to model, then integrate to forecast.

Differencing, Precisely

The single operation at the heart of ARIMA

$$\Delta X_t = X_t - X_{t-1} = (1 - L) X_t \quad (\text{first difference})$$

What it does

- Replaces each value with its change from the period before.
- In lag-operator form it is simply $(1 - L)$ applied to X_t .
- One pass removes a linear stochastic trend.

Why it works

- A random walk's differences are white noise — instantly stationary.
- Differencing a drifting series leaves a stationary series plus a constant.
- That constant becomes the drift in the forecast.

The Order of Integration, d

How many differences does the series need?

$$X_t \sim I(d) \iff (1 - L)^d X_t \text{ is stationary (ARMA)}$$

The ladder, revisited

- $d = 0$: already stationary — plain ARMA (Week 3).
- $d = 1$: one difference (most trending series, prices).
- $d = 2$: two differences (rare — some accelerating series).

Choosing d well

- d is almost always 0, 1, or 2 — seldom more.
- Under-differencing leaves a trend; over-differencing adds noise.
- Tests and the ACF (a few slides on) pin it down.

ARIMA(p, d, q)

Three integers describe the whole model

$$\underbrace{\phi(L)}_{\text{AR } p} \underbrace{(1 - L)^d}_{\text{I } d} X_t = c + \underbrace{\theta(L)}_{\text{MA } q} \varepsilon_t \quad \text{ARIMA}(p, d, q)$$

Reading the three numbers

- p: the AR order — memory of past levels.
- d: the number of differences to reach stationarity.
- q: the MA order — memory of past shocks.

How it is fit

- Difference the series d times to make it stationary.
- Identify and estimate an ARMA(p, q) on the result.
- The software does all three steps in one call.

The Random Walk Is ARIMA(0,1,0)

Old friends, in the new language

$$(1 - L) X_t = \varepsilon_t \iff \text{random walk} = \text{ARIMA}(0, 1, 0)$$

The special cases

- ARIMA(0,1,0): a random walk — difference once, model nothing.
- ARIMA(0,1,0) with a constant: a random walk WITH drift.
- ARIMA(p,0,q): an ordinary stationary ARMA.

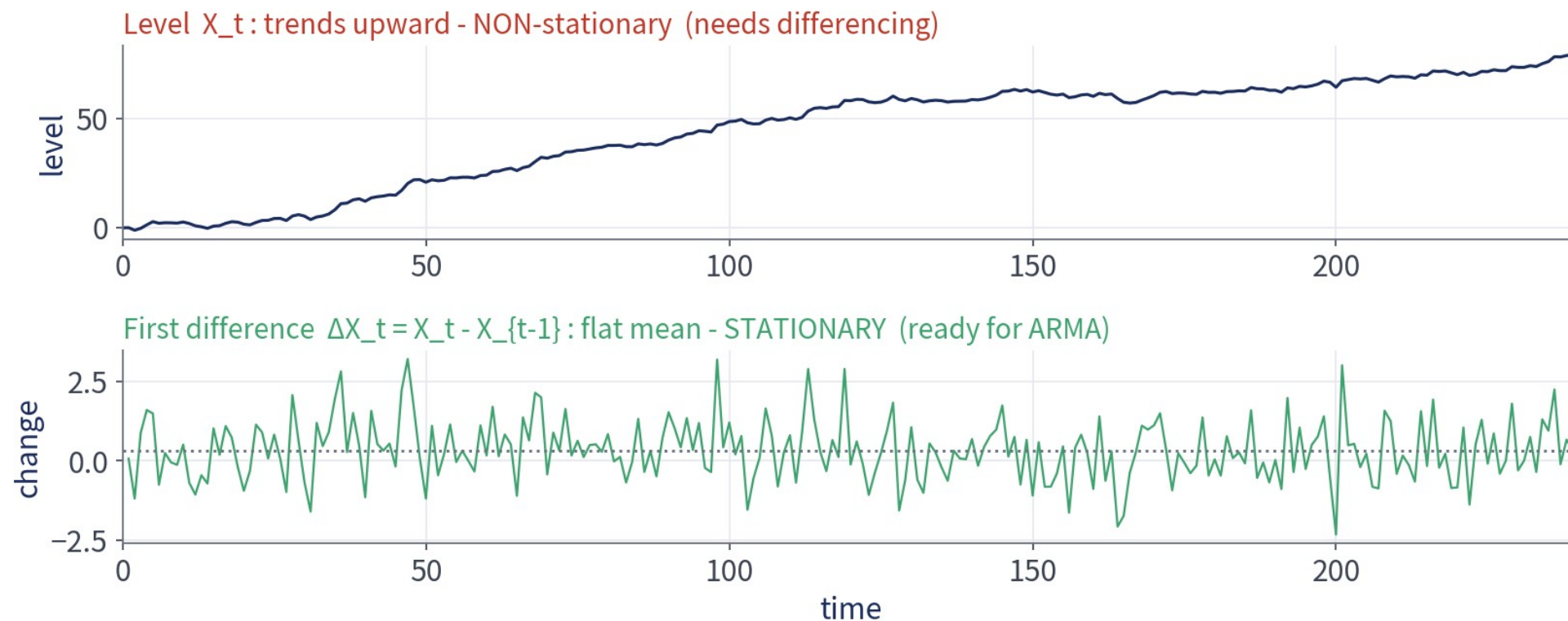
Why this is satisfying

- The unit-root series of Week 2 now has a name and a slot.
- ARIMA unifies random walks, trends, and ARMA in one family.
- Every model this course has met is an ARIMA special case.

One Difference Turns Trend Into Stationarity

Top: a trending, non-stationary level. Bottom: its first difference fluctuates around a flat mean — exactly the stationary input ARMA requires.

The 'I' in ARIMA: one difference turns a trending series into a stationary one



When Should You Difference?

Reading the series for a stochastic trend

Difference when the series has a stochastic trend — a wandering level with no fixed mean and a slowly-decaying ACF. That is the unit-root signature from Week 2.

Signs you need $d \geq 1$

- The level wanders with no mean it returns to.
- The ACF decays very slowly, staying near 1.
- ADF fails to reject a unit root; KPSS rejects stationarity.

Signs you do NOT

- The series already fluctuates around a stable mean.
- Its ACF dies out quickly.
- Differencing here would over-difference (adds noise).

How Many Differences? Let the Tests Decide

Reusing the Week-2 toolkit

The procedure

- Run ADF and KPSS on the level; if non-stationary, difference once.
- Re-test the differenced series; repeat only if still non-stationary.
- Stop as soon as the tests agree it is stationary — that is d .

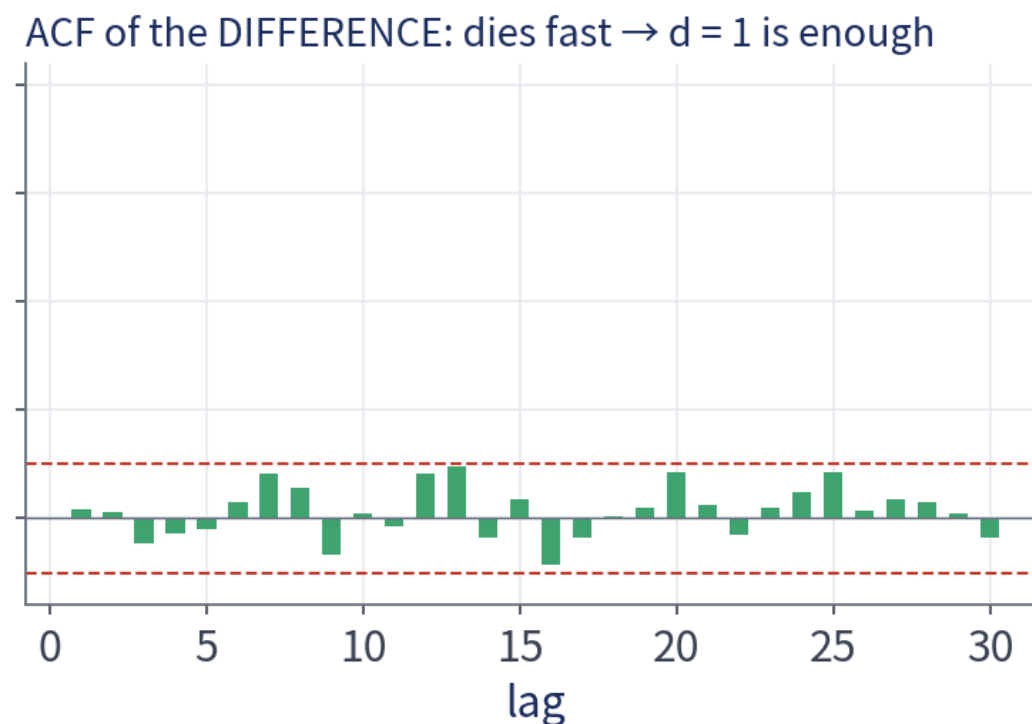
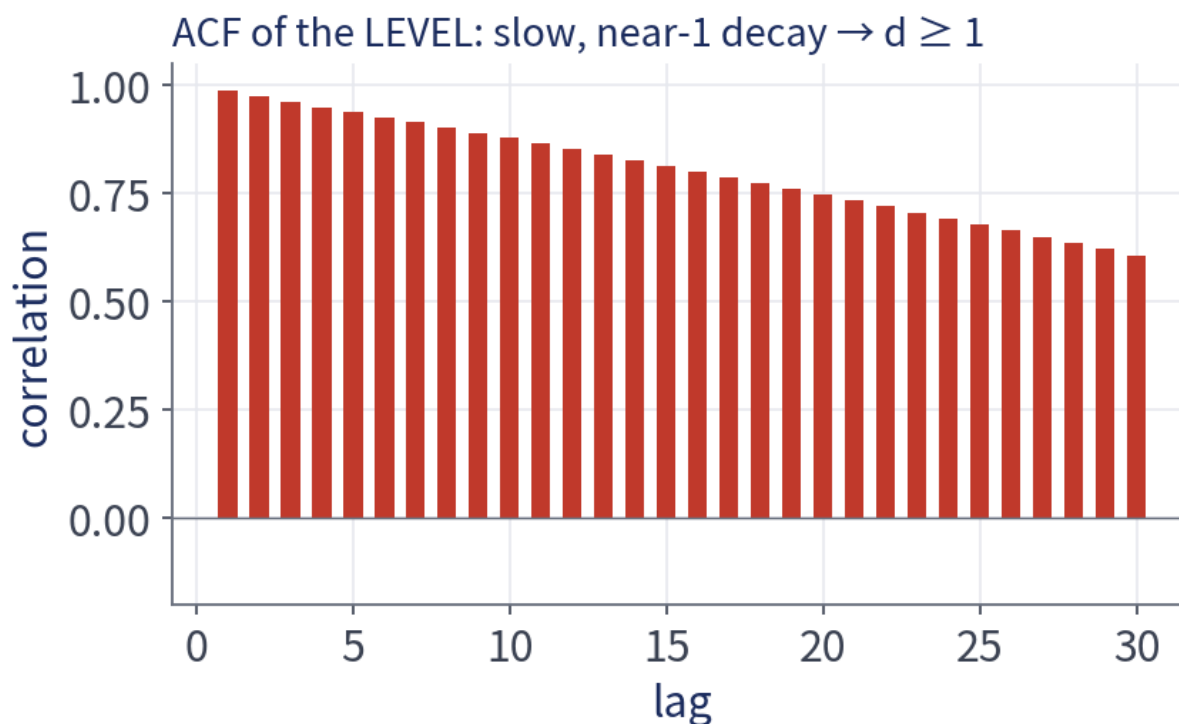
Practical guidance

- Almost every financial or macro series needs $d = 0$ or $d = 1$.
- $d = 2$ is uncommon; $d \geq 3$ almost never justified.
- When unsure, prefer the smaller d and check the residuals.

Choosing d From the ACF

A slowly-decaying ACF on the level (left) says 'difference'. After one difference the ACF dies out fast (right) — $d = 1$ is enough; a further difference would over-do it.

Choosing d: difference until the ACF stops decaying slowly



Over-Differencing, Again

Too much of a good thing

Differencing a series that is already stationary injects a large negative lag-1 autocorrelation and inflates the variance — the tell-tale sign you have differenced once too many.

The symptoms

- A strong negative spike at lag 1 in the differenced ACF.
- Noticeably larger variance than the original series.
- An MA term that keeps trying to undo the extra difference.

The remedy

- Choose the smallest d that achieves stationarity.
- Confirm with ADF/KPSS rather than differencing on reflex.
- If in doubt, difference less, not more.

Difference vs Detrend — Still Matters

ARIMA is for stochastic trends, not deterministic ones

Difference-stationary (use ARIMA)

- A stochastic trend — a unit root, random-walk-like.
- Shocks are permanent; the level wanders.
- Cure: difference (this is the 'I' in ARIMA).

Trend-stationary (detrend instead)

- A fixed deterministic trend, $a + b \cdot t$.
- Shocks are temporary; the series hugs the line.
- Cure: subtract the trend, then model the residual with ARMA.

Modeling the Differenced Series

Once stationary, it is just ARMA

After differencing d times, the hard part is over: identify p and q on the differenced series using the Week-3 ACF/PACF cheat sheet, exactly as before.

What carries over

- The ACF/PACF fingerprints identify p and q as usual.
- AIC/BIC still select among candidates.
- Residual diagnostics still gate the final model.

The only new step

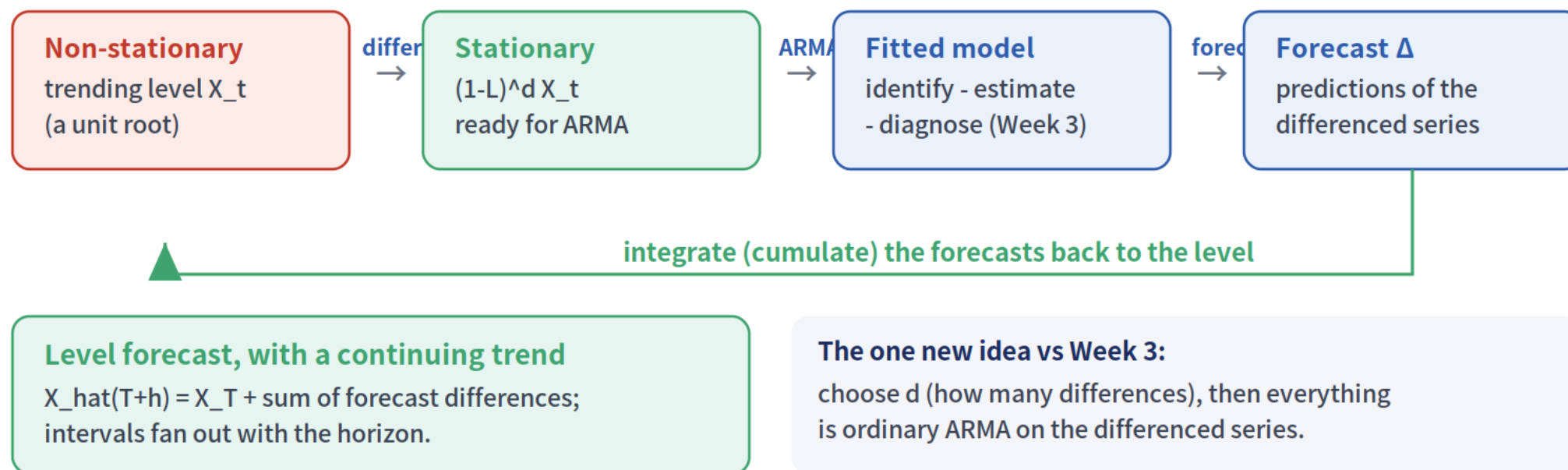
- You chose d first, and you will integrate the forecast back.
- Everything between is ordinary ARMA on the difference.
- So Week 3's entire toolbox applies unchanged.

The ARIMA Pipeline

Difference until stationary, fit an ARMA, forecast the differences, then integrate those forecasts back up to the level — trend and all.

The ARIMA Pipeline

ARIMA = difference until stationary, fit an ARMA, then integrate the forecast back to levels.



Building an ARIMA, Step by Step

Five steps from raw level to forecast

1 · Test

pick d

ADF/KPSS on
the level;
difference until
stationary.

2 · Identify

p and q

Read the
ACF/PACF of the
differenced
series.

3 · Estimate

fit

Fit
ARIMA(p,d,q);
compare by
AIC/BIC.

4 · Diagnose

residuals

Confirm the
residuals are
white noise.

5 · Forecast

integrate

Predict, then
cumulate back
to the level.

Only steps 1 and 5 are new; steps 2–4 are the Week-3 loop, unchanged.

Forecasting: Integrate the Differences Back

The trend continues, not reverts

$$\hat{X}_{T+h} = X_T + \sum_{i=1}^h \widehat{\Delta X}_{T+i} \quad (\text{integrate the forecasts back})$$

How it works

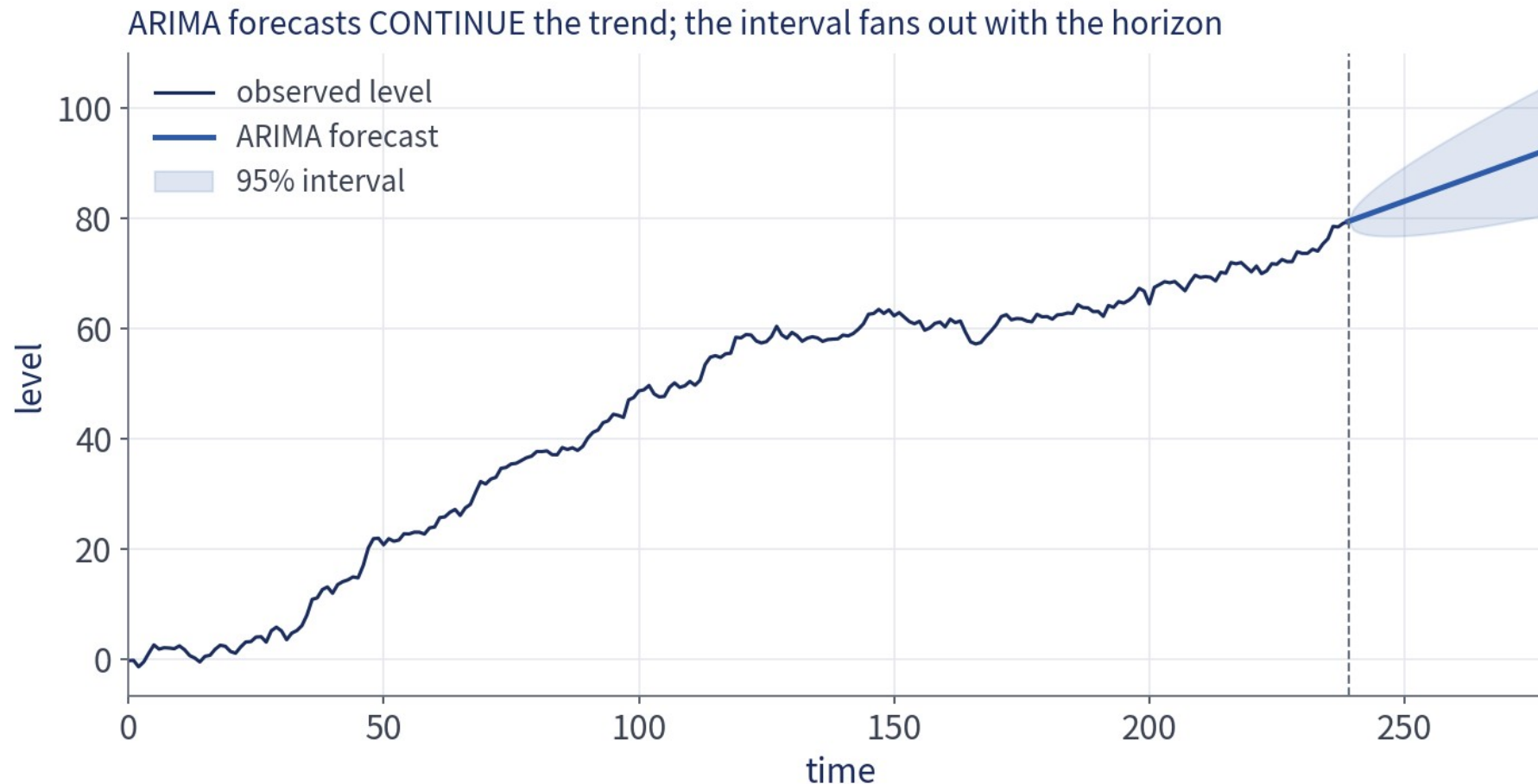
- The model forecasts the DIFFERENCES (the stationary part).
- Cumulate them onto the last observed level to get level forecasts.
- Any drift term makes the forecast keep trending.

How it differs from ARMA

- A stationary ARMA forecast reverts to its mean (Week 3).
- An ARIMA forecast instead CONTINUES the trend.
- The intervals fan out faster, reflecting the unit root.

An ARIMA Forecast Continues the Trend

Unlike a stationary ARMA, the point forecast extends the trend from the last level, and the interval fans out with the horizon — the footprint of the unit root.



ARIMA vs a Deterministic-Trend Model

Two ways to handle a trend — with different forecasts

Approach	Trend it assumes	How to handle it	Long-horizon forecast
ARIMA (difference)	stochastic (unit root)	difference d times	continues from the last level; wide bands
Detrend + ARMA	deterministic line $a+b \cdot t$	subtract the fitted trend	reverts to the line; narrower bands

Choosing wrong biases every long-horizon forecast — the same lesson as Week 2, now for forecasting.

Difference to Stationarity, Then Fit ARIMA

Direct Claude Code through the choose-d workflow

GOAL Take a trending series, use tests to choose d , fit an ARIMA, and forecast the level with honest intervals.

[INPUT] Prompt · Claude Code

Write and run `arima_lab.py` that:

1. downloads a trending series (e.g. a stock PRICE level, or airline-passenger data);
 2. runs ADF/KPSS on the level and on the first difference to choose d ;
 3. identifies (p, q) from the differenced ACF/PACF, fits $\text{ARIMA}(p, d, q)$, and reports AIC/BIC and the residual Ljung-Box;
 4. forecasts 20 steps ahead on the LEVEL with a 95% interval, and plots it.
- Tell me the chosen (p, d, q) and why.

• Expected $d = 1$ for most price levels.

- The forecast should continue the trend, not flatten.
- Compare it to a plain ARMA on the level — see the difference.
- Keep it; Part 2 adds the seasonal terms.

Deliverable: `arima_lab.py` + a level forecast with its fanning interval.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1	A price level is $I(1)$. What are its p , d , q if the difference is white noise?	(0, 1, 0) random walk
Q2	Your differenced series shows a big negative lag-1 ACF. What went wrong?	Over-differenced
Q3	How does an ARIMA forecast of a trend differ from an ARMA forecast?	Continues, not reverts

All three follow from differencing, the order d , and integrating the forecast back.

Three Pitfalls to Avoid

Where ARIMA modeling goes wrong

Common mistakes

- "When in doubt, difference more."
- "ARIMA and a deterministic trend give the same forecast."
- "Just fit ARMA to the price level directly."

The reality

- Over-differencing injects a negative lag-1 ACF and inflates variance.
- Stochastic and deterministic trends forecast very differently.
- Fitting ARMA to a unit-root level is the spurious trap (Week 2).

Three Things to Carry Forward

Part 1 in one screen

1

ARIMA = difference, then ARMA

The 'I' is just an integer d : difference the series d times to reach stationarity, then apply the entire Week-3 ARMA toolbox to the result.

2

Choose d with tests, not reflex

Use ADF/KPSS and the ACF to pick the smallest d that yields stationarity. Under-differencing leaves a trend; over-differencing adds noise.

3

ARIMA forecasts continue the trend

Forecast the differences, then integrate back: the point forecast extends the trend and the interval fans out — the signature of a unit root.

Next — Part 2: Seasonality and SARIMA, adding the repeating yearly rhythm the airline data showed.